# Poke-RL

## Does move variance change how PPO learns Pokémon battles? (Midsem)

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# Demonstration

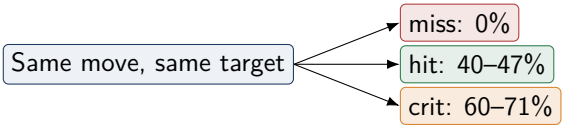
# Pokémon battles in one slide

## The game (singles)

- Each side has a team (here **4**); one Pokémon is active
- Every turn **both players choose at once**: one of 4 moves, or switch
- The faster Pokémon acts first
- You win when all opposing Pokémon have fainted

## Randomness in every attack

- **Accuracy**: Hydro Pump hits 80% of the time
- **Damage roll**: uniformly 85–100% of the maximum
- **Critical hit**: 1 in 24 chance, $\times 1.5$ damage
- **Secondary effects**: e.g. 10% burn, 30% flinch




(illustrative numbers)

# Why reinforcement learning?

## The problem fits RL

- No labelled "correct move": only **win or lose** at the end
- **Sequential**: a Swords Dance now pays off turns later
- **Stochastic**: good play means good *on average*

**Formalised as an MDP $\langle S, A, P, R, \gamma \rangle$**

states $S$ · actions $A$ · transition probabilities $P(s' \mid s, a)$ (dice + opponent) · reward $R$ · discount $\gamma$

**Too many states to enumerate**

HP alone: ∼300 values per Pokémon, 8 Pokémon

$$300^8 \approx 7 \times 10^{19} \text{ states}$$

before status, stat boosts ($13^7$ per active), hazards, PP…

**So: deep RL**

A table with one entry per state is impossible. A **neural network** maps the state (763 numbers) to move probabilities and **generalises** across similar states.

# Technically a POMDP

## What is a POMDP?

A **partially observable** MDP $\langle S, A, P, R, \Omega, O, \gamma \rangle$: the agent never sees the true state $s$, only an observation

$$o \sim O(o \mid s)$$

So the best policy may need the whole history (a *belief* over states).

## Why Pokémon is one

- Unrevealed opposing Pokémon, moves, items, stats
- **Simultaneous moves**: the opponent's choice this turn is unknown
- Hidden counters (e.g. sleep turns left)

## Why we treat it as (nearly) an MDP

1. **Open team sheets** reveal every set from turn 1: the largest hidden part is gone
2. **Dice are not hidden state**: they are random transitions $P(s' \mid s, a)$, which an MDP already models
3. The heuristic opponent is **(nearly) deterministic** given the battle, so its choice behaves like part of $P$

$\Rightarrow$ a memoryless policy $\pi(a \mid o)$ is a reasonable approximation.

# The question

**Long-term goal (Final)**

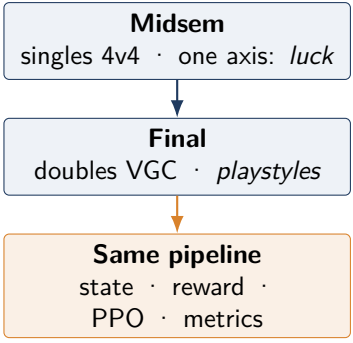
How does RL learn different **playstyles** (standard vs Perish Trap) in doubles VGC?

**Why start with variance?**

Playstyles differ in many ways at once and are very hard to measure. Variance can be changed **alone**, ceteris paribus.

**Midsem question**

Does a team that relies on **inaccurate moves** make PPO learn **slower** or **less reliably**?



# Our team and the opponents

| Pokémon | LowVar move | | HighVar move | $\mathbb{E}[\text{BP}] = \text{BP} \times \text{acc}$ | SD |
|---|---|---|---|---|---|
| Moltres | Flamethrower 90/100 | $\rightarrow$ | Fire Blast 110/85 | $90 \rightarrow 93.5$ | $0 \rightarrow 39$ |
| Moltres | Air Slash 75/95 | $\rightarrow$ | Hurricane 110/70 | $71 \rightarrow 77$ | $16 \rightarrow 50$ |
| Primarina | Surf 90/100 | $\rightarrow$ | Hydro Pump 110/80 | $90 \rightarrow 88$ | $0 \rightarrow 44$ |
| Primarina | Ice Beam 90/100 | $\rightarrow$ | Blizzard 110/70 | $90 \rightarrow 77$ | $0 \rightarrow 50$ |

Unchanged controls: **Great Tusk**, **Kingambit**. Items, abilities, spreads, other moves identical (diff-checked).

**Opponent A**

Glimmora · Dragapult · Dragonite · Iron Valiant

Heuristic with our team wins: **5–8%**

**Opponent B**

Landorus-T · Gholdengo · Zamazenta · Kyurem

Heuristic with our team wins: **98–100%**

**Opponent C**

Corviknight · Clefable · Gliscor · Raging Bolt

Heuristic with our team wins: **24–25%**

No weather or accuracy modifiers on any team. Technically: Not purely variance: Blizzard lowers the mean; Air Slash (flinch) vs Hurricane (confusion).

# Setup

| | |
|---|---|
| Format | Gen 9 OU singles, **no Tera**, 4v4, fixed lead order |
| Information | **Full open team sheets** for our agent |
| Opponent | poke-env `SimpleHeuristicsPlayer`, team A/B/C at random each battle |
| Algorithm | `MaskablePPO` (sb3-contrib 2.9.0) |
| Observation / actions | 763 features / 26 actions with legality masks |
| Budget | 200k decision steps per run ($\approx$ 15,300 battles) |
| Runs | 2 versions $\times$ **5 seeds** (0–4) $=$ 10 runs |
| Compute | CPU, 3 runs in parallel, $\approx$ 0.4 h per run |

**Why 200k steps**

A 2M-step pilot reached 100% by 30–80k steps and stayed there, so we spent compute on **seeds** instead of length.

# State and actions

## State: 763 numbers

- **8 Pokémon rows**: HP, status, active; stats, types, item, ability; 4 moves with power, **accuracy**, effectiveness, PP
- **Active matchup**: stat boosts, who moves first
- **Field**: hazards, screens, turn

## Accuracy is a feature

Otherwise the agent could tell Surf from Hydro Pump only by outcomes.

## Actions: 26, masked

| | |
|---|---|
| 0–3 | switch |
| 4–5 | never (4v4) |
| 6–9 | move 1–4 |
| 10–25 | never (Tera etc.) |

$$\pi(a \mid s) = \frac{m(a)\, e^{z_a}}{\sum_b m(b)\, e^{z_b}}$$

Illegal actions get probability 0, at sampling *and* update time.

# Reward: fixing sparse rewards

**Sparse true reward**

$+1$ win, $-1$ loss, 0 otherwise. One signal after $\sim$15 decisions.

**Potential: "how well am I doing metric" ($N = 4$)**

$$\Phi(s) = \frac{\mathrm{HP}_{\mathrm{ours}} - \mathrm{HP}_{\mathrm{opp}}}{2N} + \frac{\mathrm{alive}_{\mathrm{ours}} - \mathrm{alive}_{\mathrm{opp}}}{2N}$$

**Shaped reward**

$$r'_{t+1} = r_{t+1} + \gamma\,\Phi(s_{t+1}) - \Phi(s_t)$$

with $\Phi(s_T) = 0$ at the end of the battle

**Same goal, earlier feedback**

$$\sum_t \gamma^t r'_{t+1} = \sum_t \gamma^t r_{t+1} + \underbrace{\gamma^T \Phi(s_T)}_{0} - \underbrace{\Phi(s_0)}_{0}$$

Shaped return = true return in **every** battle, so the best policy is unchanged.

# Training with PPO

**Actor-critic**

Actor $\pi_\theta(a \mid s)$ picks moves · critic $V_\phi(s)$ predicts the return

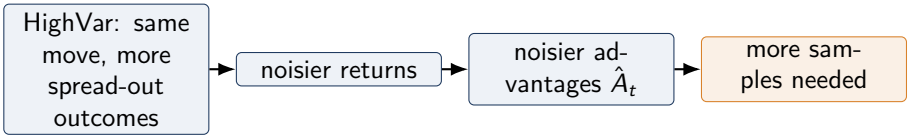
$$\hat{A}_t = \sum_l (\gamma\lambda)^l \delta_{t+l}, \quad \delta_t = r'_{t+1} + \gamma V(s_{t+1}) - V(s_t)$$

$$L^{\text{CLIP}} = \mathbb{E}\Big[\min\big(\rho_t \hat{A}_t,\ \text{clip}(\rho_t, 0.8, 1.2)\hat{A}_t\big)\Big]$$

$\hat{A} > 0$: "better than expected, do it more". Clipping stops one noisy batch moving the policy far.

| Setting | Value |
|---|---|
| Network | 256-256 MLP |
| Learning rate | $3 \times 10^{-4}$ |
| Rollout / batch | 2048 / 128 |
| Epochs per update | 10 |
| $\gamma$ / $\lambda$ | 0.99 / 0.95 |
| Clip / entropy | 0.2 / 0.01 |

# Hypotheses




- **H1 Speed**: HighVar learns slower (lower AUC, later convergence)
- **H2 Reliability**: HighVar seeds disagree more [Henderson et al. 2018]
- **H3 Strength**: similar once learned (mean power $\approx$ equal)

**Why the effect may be small**

PPO's critic baseline, GAE and clipping exist to reduce gradient noise [Schulman et al. 2016, 2017]. For PPO, noise adds *variance*, not *bias*.

# What we measure

| Metric | Definition | How measured |
|---|---|---|
| Eval win rate $w(k)$ | Greedy policy's win rate vs the heuristic at step $k$ | Every 5k steps: 120 battles, 40 per opponent team; Wilson 95% CI |
| **AUC** (speed) | Average of $w(k)$ over training, $\frac{1}{K}\int_0^K w(k)\,dk$ | 0–200k (primary) and 0–50k (learning phase) |
| Steps to converge | First step where $w \geq 0.95$ for 3 evals in a row | From the eval curve |
| Seed spread | Std of a metric over the 5 seeds | Per version |
| Stability | Evals below 0.95 after 60k; largest drop between evals | From the eval curve |
| Final strength | Win rate of the final agent | 1,020 battles vs heuristic; 204 vs Random and MaxBasePower |
| RL lift | Final win rate − heuristic baseline (same team) | Baseline: 1,500 battles |
| Significance | LowVar vs HighVar difference | Exact permutation test over all 252 splits |

# Baselines: the heuristic piloting our team

| | LowVar | HighVar |
|---|---|---|
| Win rate (1,500 battles) | 0.429 | 0.437 |
| 95% CI | [0.404, 0.454] | [0.412, 0.463] |
| vs A | 0.05–0.08 | |
| vs B | 0.98–1.00 | |
| vs C | 0.24–0.25 | |
| Swap-move miss rate | **2%** | **23%** |

**What it tells us**

- The manipulation is **active**: $\sim 10\times$ more misses on HighVar
- . . . but win rate is **the same** (CIs overlap)
- The **matchup** decides the game, not the swapped moves
- Sets the reference for RL lift ($\approx 0.43$)

# Result: PPO learns to beat the heuristic

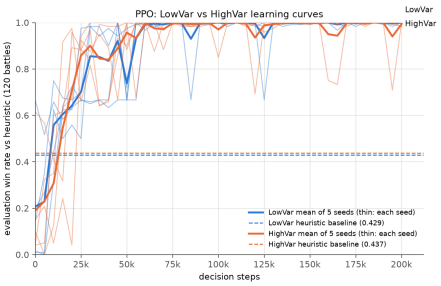



Figure: Eval win rate vs steps: 5 seeds per version

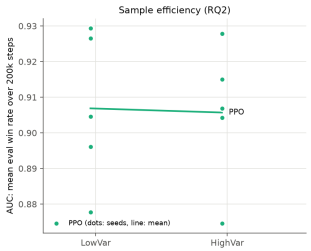
- Untrained: ≈0.20
- Converges by **50–75k** steps
- Plateau at ≈0.67: A and B beaten, C not yet
- Every run learns **A** → **B** → **C**
- LowVar and HighVar curves **overlap**

# Result: speed and seed spread (H1, H2)

| Mean ± std, 5 seeds | LowVar | HighVar | $p$ |
|---|---|---|---|
| AUC 0–200k | 0.907 ± 0.022 | 0.906 ± 0.020 | 0.91 |
| AUC 0–50k | 0.667 ± 0.090 | 0.670 ± 0.075 | 0.96 |
| Steps to $\geq$0.95 | 50k ± 12k | 52k ± 20k | 0.95 |

**Reading**

The AUC gap ($-0.001$) is $\sim 20\times$ smaller than the seed spread. Seed-by-seed differences go both ways.



# Result: stability after convergence (exploratory)

| From 60k steps on | LowVar | HighVar | $p$ |
|---|---|---|---|
| Dips below 0.95 per run | 0.4 | **1.4** | 0.14 |
| Mean eval win rate | 0.993 | 0.987 | 0.14 |
| Largest drop (whole run) | 0.22 | 0.25 | 1.00 |

**A hint, not a result**

Same direction as the 2M pilot, but $p = 0.14$ with 5 seeds, and we chose this metric after seeing the data.

**Every dip looks the same**

- **One** opponent collapses, the others stay at 100%
- e.g. HighVar s1 at 120k: A 1.00, B 1.00, **C 0.08**
- Recovers by the next evaluation

$\Rightarrow$ the policy wins with **one narrow line per matchup**; a small update can break it.

# Result: final strength (H3)

| Final agent (mean of 5) | LowVar | HighVar | $p$ |
|---|---|---|---|
| vs heuristic (1,020) | **0.9998** | **0.9994** | ceiling |
| vs Random (204) | 0.979 | 0.965 | 0.17 |
| vs MaxBasePower (204) | 0.926 | 0.918 | 0.86 |
| RL lift vs baseline | +0.571 | +0.562 | |

- Every run $\geq 0.991$ (Wilson lower bound)
- No difference in final strength
- **Specialised**: worse vs weaker players it never met (vs A as low as 0.41)

**Interpretation**

It learned to **exploit this opponent**, not to play the matchups well in general.

# Verdict, and why

| Hypothesis | Result |
|---|---|
| H1 Speed | Not supported ($p = 0.91$) |
| H2 Reliability | Not supported (std 0.022 vs 0.020) |
| H3 Strength | Consistent, but at ceiling |
| Stability | Hint only ($p = 0.14$) |

**Most likely reasons**

1. **Ceiling**: deterministic opponent, both versions reach 100%
2. **Swapped moves may not be used**: replays show a Kingambit Swords Dance sweep with no swapped move (not yet measured)
3. Even the heuristic's win rate ignores 23% vs 2% misses

**Conclusion**

With this opponent, the variance manipulation does not reach the learner. This is **not** evidence that variance never matters.

# A possible limitation: full information

**Our choice**

Our agent sees **every set** from turn 1, including EVs, IVs and nature. The heuristic sees only what is revealed.

**Why we chose it**

- Removes hidden information, so the remaining noise is the **dice** we manipulate
- Makes the problem nearly an MDP

**What it may cost**

- **Asymmetric**: part of the +0.57 lift is information, not learning
- May make the opponent **easier to exploit**, contributing to the ceiling
- Stronger than official VGC open team sheets, which hide spreads

**Why the comparison still holds:** the advantage is identical for LowVar and HighVar.

Future: official-OTS and no-OTS variants.

# Other limitations

**Statistics**

- 5 seeds per version: only large effects detectable [Agarwal et al. 2021]
- 120 battles per eval: $\pm 0.09$ at a 50% win rate
- Stability and per-opponent analyses chosen after seeing data

**Design**

- One deterministic, exploitable opponent
- Swap-move usage by the agent not measured
- Not purely variance: Blizzard lowers the mean; 110 BP changes KO thresholds
- Near-Markov only (simultaneous moves)

# Next steps

## Fix the midsem experiment

1. **Measure** swap-move usage and misses of the trained agents
2. **Remove the ceiling**: stochastic or stronger opponents (Metamon, Foul Play), held-out team
3. Teams where the risky moves are **needed** to win
4. Add **DQN**: its max-based target should turn noise into bias

## Final: same spec, doubles

| | Midsem | Final |
|---|---|---|
| Format | singles 4v4 | doubles VGC |
| Axis | luck | playstyle |
| Actions | 26 | per slot $\times$ target |
| Reward | $\pm 1$ + shaping | same |
| Algorithms | PPO | PPO, DQN |
| Opponents | heuristic | + existing bots |
| Metrics | AUC, stability, lift | same |

# References


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# Backup slides

# Backup: shaping, worked example

3-step battle we win, $\gamma = 0.99$:

| Step | $\Phi$ before | $\Phi$ after | $r'$ |
|---|---|---|---|
| 1 | 0 | 0.0625 | 0.0619 |
| 2 | 0.0625 | 0.25 | 0.1850 |
| 3 (win) | 0.25 | **0** | 0.7500 |

$$0.0619 + 0.99(0.1850) + 0.99^2(0.75) = 0.9801 = \gamma^2 \cdot 1$$

**Two bugs we avoided**

- $\Phi(s_T) \neq 0$: extra $\gamma^T \Phi(s_T)$ pays for winning big and fast
- poke-env's helper omits $\gamma$: small bonus for being ahead every step

The critic learns $V(s) - \Phi(s)$; the shift is equal for all actions, so the best action is unchanged.

# Backup: discount and battle length

**Why $\gamma = 0.99$**

$$0.99^{15} \approx 0.86, \qquad \frac{1}{1-\gamma} = 100 \text{ steps}$$

The win signal reaches the first move almost undiscounted. GAE credit window $\frac{1}{1-\gamma\lambda} \approx 17$ steps $\approx$ one battle. Not tuned.

**Steps vs turns**

A step is one decision. Forced switches add steps without turns.

- Early training: ≈18 steps per battle
- Last 50k: 10.5–14 steps (9.5–12.4 turns)
- Heuristic baseline: 10.4 turns
- No battle hit the 300-step cap

Training battle length per run

`figures/ep_len.png`

# Backup: PPO details

**Clipping in numbers**

$\pi_{\text{old}}(a) = 0.5$, $\hat{A} > 0$. An update to 0.65 gives $\rho = 1.3$, clipped to 1.2: no gradient beyond $\pi \approx 0.6$.

**GAE $\lambda$**

$\lambda = 0$: trust the critic (low variance, biased). $\lambda = 1$: full returns (unbiased, noisy). 0.95 is the standard compromise.

**Full loss**

$$L = -L^{\text{CLIP}} + c_v \, \mathbb{E}[(V_\phi - \hat{R})^2] - c_e \, \mathbb{E}[\mathcal{H}(\pi)]$$

**Masking at update time**

The ratio $\rho$ must use the same masked distribution as sampling; MaskablePPO stores the masks.

# Backup: statistics

**Exact permutation test**

Pool the 10 runs, try all $\binom{10}{5} = 252$ ways to split them into two groups of 5, and count how often the split difference is at least as large as the real one.

No distribution assumption. Smallest possible $p = 2/252 \approx 0.008$.

**Wilson 95% interval**

$$\frac{\hat{p} + \frac{z^2}{2N} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}}}{1 + \frac{z^2}{N}}, \quad z = 1.96$$

$N = 120$: $\pm 0.09$ at 50%; at 100% the lower bound is 0.97.
$N = 1{,}020$: much tighter.

# Backup: learning order by opponent

Steps until each opponent is beaten at $\geq$0.95 for 2 evals in a row (seeds 0–4)

| | A | B | C |
|---|---|---|---|
| LowVar | 10k, 10k, 10k, 10k, 25k | 30k, 25k, 25k, 5k, 25k | 30k, 55k, 55k, 30k, 10k |
| HighVar | 20k, 20k, 15k, 20k, 40k | 35k, 50k, 20k, 15k, 25k | 35k, 55k, 50k, 30k, 60k |

- Always A first, C last: C is the bottleneck
- A is the reverse of the baseline (heuristic wins only 5–8% vs A)
- HighVar beats A later (23k vs 13k), $p = 0.13$, exploratory

# Backup: the late dips

| Run | Step | vs A | vs B | vs C |
|---|---|---|---|---|
| LowVar s1 | 125k | 1.00 | 1.00 | 0.00 |
| LowVar s3 | 85k | 1.00 | 1.00 | 0.00 |
| HighVar s1 | 65k / 120k | 1.00 | 1.00 | 0.63 / 0.08 |
| HighVar s2 | 100k / 195k | 1.00 / 0.13 | 1.00 | 0.55 / 1.00 |
| HighVar s3 | 160k, 165k | 1.00 | 0.25, 0.20 | 1.00 |
| HighVar s4 | 70k | 1.00 | 1.00 | 0.65 |

One run's curve with a dip

`figures/example_dip_run.png`

# Backup: is 100% suspicious? Sanity checks

- Untrained agent: ≈0.20; heuristic pilot: 0.43
- Learning is gradual (A → B → C), not instant
- Shaping sum = 0 in every battle
- No battle hit the step cap
- Watched replays: legal, sensible play

**Why it is plausible**

A near-deterministic heuristic plays the same way in the same position, and our agent sees full sets. A fixed winning line exists and PPO finds it.

Worse results vs Random and MaxBasePower show **specialisation**, not a bug.

# Backup: why only PPO? (and the DQN prediction)

**DQN target**

$$y = r' + \gamma \max_{a'} Q_{\bar{\theta}}(s', a')$$

$\mathbb{E}[\max \hat{Q}] \geq \max \mathbb{E}[\hat{Q}]$: noise becomes **upward bias** that compounds [Thrun & Schwartz 1993; van Hasselt 2010].

- PPO's critic learns $V$ with **no max**: noise $\rightarrow$ variance only
- Prediction: DQN is *more* hurt by HighVar than PPO
- DQN arm not ready for the midsem; it is part of the final

# Backup: pilot and the threshold metric

**2M-step pilot (3 runs)**

All reached 100% by 30–80k steps and stayed there to $\sim$870k. HighVar s0 dipped to 0.65 (30k) and 0.79 (70k), both vs C. $\Rightarrow$ budget cut to 200k, more seeds.

**Why "baseline + 0.10" was dropped**

Threshold $\approx$0.53, crossed by every run within 30k steps. Seed 1's untrained network already scores 0.62–0.67, so it "converged" at step 0. Replaced by sustained $\geq$0.95.

# Backup: state encoding in detail

| Block | Contents | Source |
|---|---|---|
| 8 Pokémon rows | HP, fainted, status, active | battle |
| | actual stats, types, item, ability | team sheet |
| | 4 moves: power, **accuracy**, category, priority, effectiveness vs opposing active, STAB, PP | sheet + battle |
| Active matchup | stat boosts, speed order, toxic counter | battle + sheet |
| Field | hazards, screens, turn | battle |
| Total | 763 features, scaled to $[0, 1]$ ($[-1, 1]$); rows keyed by species | |

**Deliberately excluded**

KO-probability features: they would compute power $\times$ accuracy *for* the agent.